

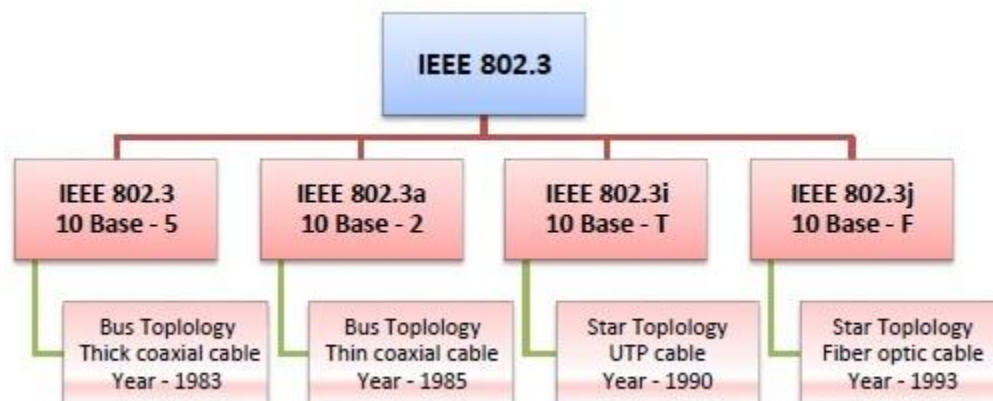
IEEE standard 802.3 & 802.11 for LAN and WLAN

IEEE 802.3 is a set of standards and protocols that define Ethernet-based networks. Ethernet technologies are primarily used in LANs, though they can also be used in MANs and even WANs. IEEE 802.3 defines the physical layer and the medium access control (MAC) sub-layer of the data link layer for wired Ethernet networks.

IEEE 802.3 Popular Versions

There are a number of versions of IEEE 802.3 protocol. The most popular ones are.

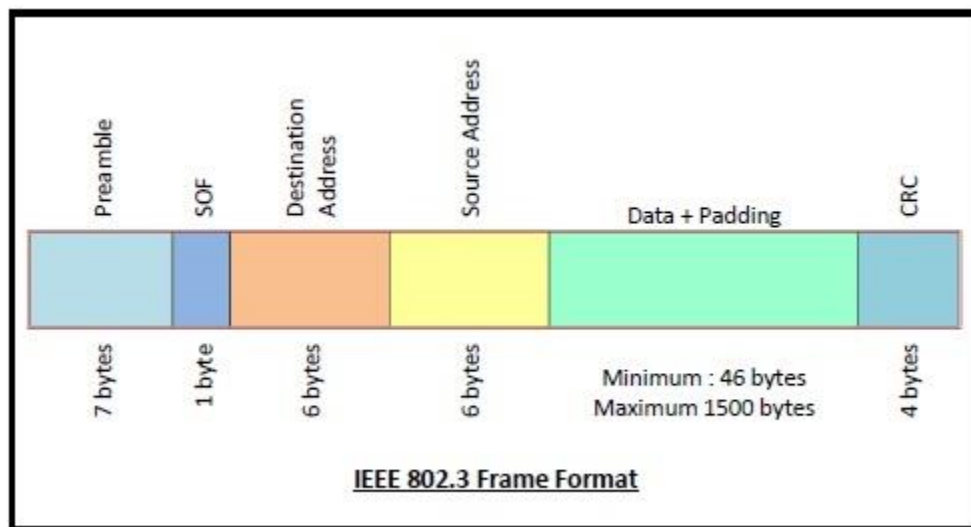
- **IEEE 802.3:** This was the original standard given for 10BASE-5. It used a thick single coaxial cable into which a connection can be tapped by drilling into the cable to the core. Here, 10 is the maximum throughput, i.e. 10 Mbps, BASE denoted use of baseband transmission, and 5 refers to the maximum segment length of 500m.
- **IEEE 802.3a:** This gave the standard for thin coax (10BASE-2), which is a thinner variety where the segments of coaxial cables are connected by BNC connectors. The 2 refers to the maximum segment length of about 200m (185m to be precise).
- **IEEE 802.3i:** This gave the standard for twisted pair (10BASE-T) that uses unshielded twisted pair (UTP) copper wires as physical layer medium. The further variations were given by IEEE 802.3u for 100BASE-TX, 100BASE-T4 and 100BASE-FX.
- **IEEE 802.3j:** This gave the standard for Ethernet over Fiber (10BASE-F) that uses fiber optic cables as medium of transmission.



Frame Format of IEEE 802.3

The main fields of a frame of classic Ethernet are -

- **Preamble:** It is a 7 bytes starting field that provides alert and timing pulse for transmission.
- **Start of Frame Delimiter:** It is a 1 byte field that contains an alternating pattern of ones and zeros ending with two ones.
- **Destination Address:** It is a 6 byte field containing physical address of destination stations.
- **Source Address:** It is a 6 byte field containing the physical address of the sending station.
- **Length:** It a 7 bytes field that stores the number of bytes in the data field.
- **Data:** This is a variable sized field carries the data from the upper layers. The maximum size of data field is 1500 bytes.
- **Padding:** This is added to the data to bring its length to the minimum requirement of 46 bytes.
- **CRC:** CRC stands for cyclic redundancy check. It contains the error detection information.



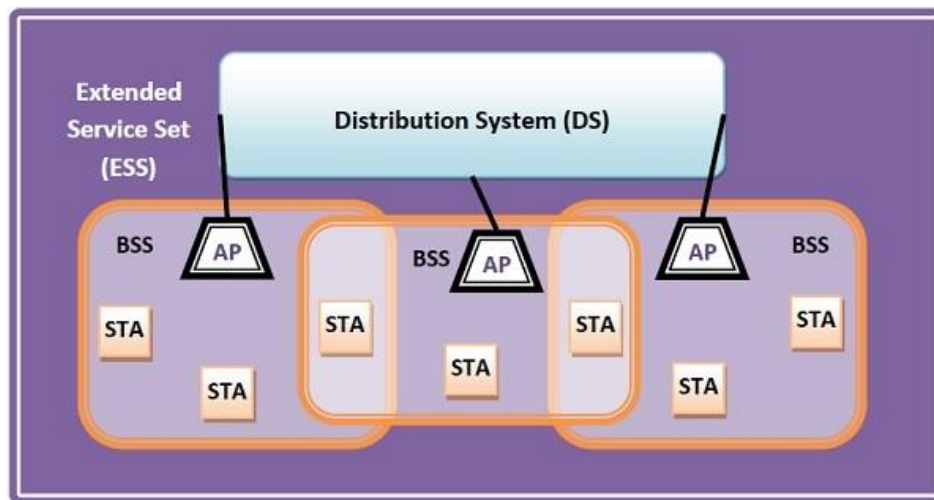
IEEE 802.11 networks

IEEE 802.11 standard, popularly known as WiFi, lays down the architecture and specifications of wireless LANs (WLANs). WiFi or WLAN uses high-frequency radio waves instead of cables for connecting the devices in LAN. Users connected by WLANs can move around within the area of network coverage.

IEEE 802.11 Architecture

The components of an IEEE 802.11 architecture are as follows –

- **Stations (STA)** – Stations comprises of all devices and equipment that are connected to the wireless LAN. A station can be of two types–
 - Wireless Access Point (WAP) – WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.
 - Client. Clients are workstations, computers, laptops, printers, smartphones, etc.
- Each station has a wireless network interface controller.
- **Basic Service Set (BSS)** – A basic service set is a group of stations communicating at the physical layer level. BSS can be of two categories depending upon the mode of operation–
 - Infrastructure BSS – Here, the devices communicate with other devices through access points.
 - Independent BSS – Here, the devices communicate in a peer-to-peer basis in an ad hoc manner.
- **Extended Service Set (ESS)** – It is a set of all connected BSS.
- **Distribution System (DS)** – It connects access points in ESS.



Frame Format of IEEE 802.11

The main fields of a frame of wireless LANs as laid down by IEEE 802.11 are –

- **Frame Control** – It is a 2 bytes starting field composed of 11 subfields. It contains control information of the frame.
- **Duration** – It is a 2-byte field that specifies the time period for which the frame and its acknowledgment occupy the channel.
- **Address fields** – There are three 6-byte address fields containing addresses of source, immediate destination, and final endpoint respectively.
- **Sequence** – It is a 2 bytes field that stores the frame numbers.
- **Data** – This is a variable-sized field that carries the data from the upper layers. The maximum size of the data field is 2312 bytes.
- **Check Sequence** – It is a 4-byte field containing error detection information.

